COURSE TITLE	**GENERATIVE AI**
COURSE CODE	**04BM0205**
COURSE CREDITS	**3**

Course Outcomes: After completion of this course, student will be able to:

1. Evaluate the fundamentals and applications of generative AI in various industries and business functions.
2. Apply generative AI tools and techniques to create innovative solutions in business scenarios.
3. Analyze the ethical, legal, and economic implications of generative AI for organizations.
4. Design AI-driven strategies to solve complex business problems.
5. Critique the role of generative AI in enhancing competitive advantage in the global business landscape.

Pre-requisite of course: Basic knowledge of AI and machine learning, Familiarity with Python programming, Understanding of basic business functions, Basic statistics and data analysis skills.

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
3	0	0	0	20	30	50	0

Contents : Unit	Topics	Contact Hours
1	**Overview of Artificial Intelligence and Machine Learning** Introduction to Generative AI: Concepts and Techniques, Types of Generative AI Models (e.g., GANs, VAEs, Transformers), Applications of Generative AI in Business, Future Trends in Generative AI, Software: Artbreeder , Case Study: Using Generative AI for Fashion Design Innovation with Artbreeder	10
2	**Tools and Technologies for Generative AI** Introduction to AI Frameworks and Tools (TensorFlow, PyTorch, OpenAI API), Creating Text-Based Content with Generative Models, Image and Video Generation using GANs, AI Tools for Data Synthesis and Augmentation, Practical Applications in Automation and Personalization, Software: ChatGPT API / OpenAI GPT-4, Case Study: How eCommerce Brands Use ChatGPT to Automate Customer Interaction	10

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Contents : Unit	Topics	Contact Hours
3	**Ethical and Legal Considerations in Generative AI** Ethical Issues in AI: Bias, Fairness, and Accountability, Legal Considerations: Intellectual Property and Data Privacy, Regulatory Landscape for AI in Business, Responsible AI and Its Role in Decision Making, Governance and Transparency in AI Development, Software: DeepArt, Case Study: Legal Dispute on AI-Generated Art and Copyright Issues Using DeepArt	10
4	**Leveraging Generative AI for Business Innovation** Innovation through AI: Creating New Business Models, Generative AI in Product Development, AI-Driven Customer Insights and Personalization, Automating Business Functions Using Generative AI, Building AI-First Business Strategies, Software: Lumen5, Case Study: Lumen5 in Digital Marketing: Automating Video Content for Brand Engagement	8
5	**Future Directions and Challenges of Generative AI in Business** The Evolution of Generative AI: From Research to Real-World Application, Scaling Generative AI Solutions in Business, The Role of AI in Digital Transformation, Challenges in AI Adoption: Data Quality, Talent, and Resources, Exploring New Frontiers: Quantum Computing and AI's Future, Software: Runway ML, Case Study: Generative AI in Healthcare: The Future of Personalized Medicine and Drug Discovery	7
Total Hours		**45**

Textbook :

1 Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, 2023

2 Hands-On Generative Adversarial Networks with Keras, Rafael Valle, Packt Publishing, 2023

References:

1 Machine Learning Yearning, Machine Learning Yearning, Andrew Ng, deeplearning.ai, 2022

2 Applied Artificial Intelligence: A Handbook for Business Leaders, Applied Artificial Intelligence: A Handbook for Business Leaders, Mariya Yao, Adelyn Zhou, Marlene Jia, Topbots Inc, 2023

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation

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Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
0.00	0.00	40.00	40.00	10.00	10.00

Instructional Method:

1 Classroom lectures and discussions
2 Hands-on sessions using AI tools and software
3 Case study analysis at the end of each module

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